CAN Bus and Monitoring System Using ZigBee

Project Report

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Abstract

This project presents the design and implementation of a CAN (Controller Area Network) Bus based monitoring system integrated with ZigBee wireless communication. The system enables real-time data acquisition from multiple nodes and transmits the data wirelessly for remote monitoring. CAN Bus ensures reliable and efficient communication between microcontrollers, while ZigBee provides low-power wireless connectivity. The system is suitable for industrial automation, automotive diagnostics, and IoT-based monitoring applications.

0.0.1 Introduction

Modern industrial and automotive systems require efficient communication and monitoring solutions. CAN Bus is widely used due to its robustness and reliability in noisy environments. However, wired communication limits scalability and flexibility.

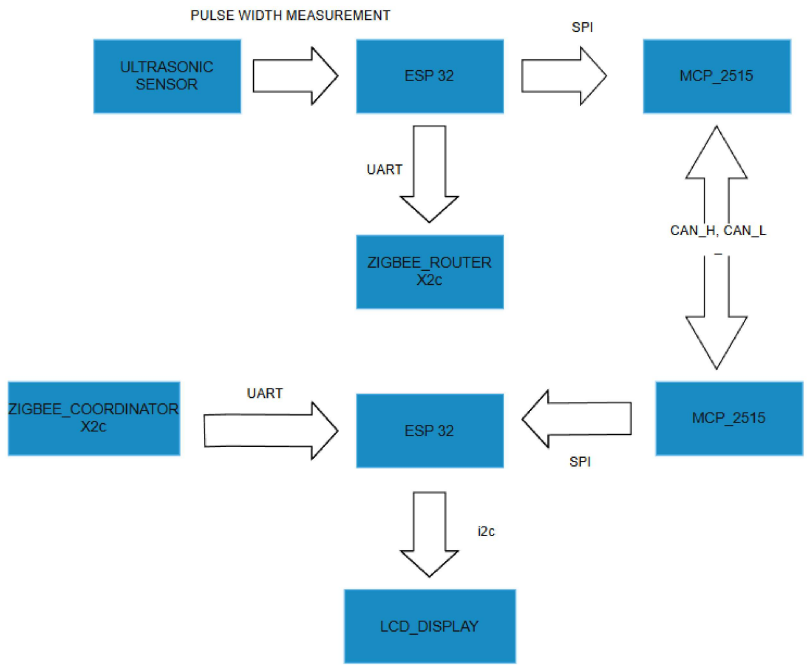
To overcome this, ZigBee wireless communication is integrated with CAN Bus, allowing remote monitoring and control. This project combines both technologies to develop a hybrid communication system.

0.1 Objectives

- To design a CAN Bus communication system
- To integrate ZigBee for wireless monitoring
- To implement real-time data acquisition
- To ensure reliable and low-power communication

1 System Overview

1.1 Block Diagram

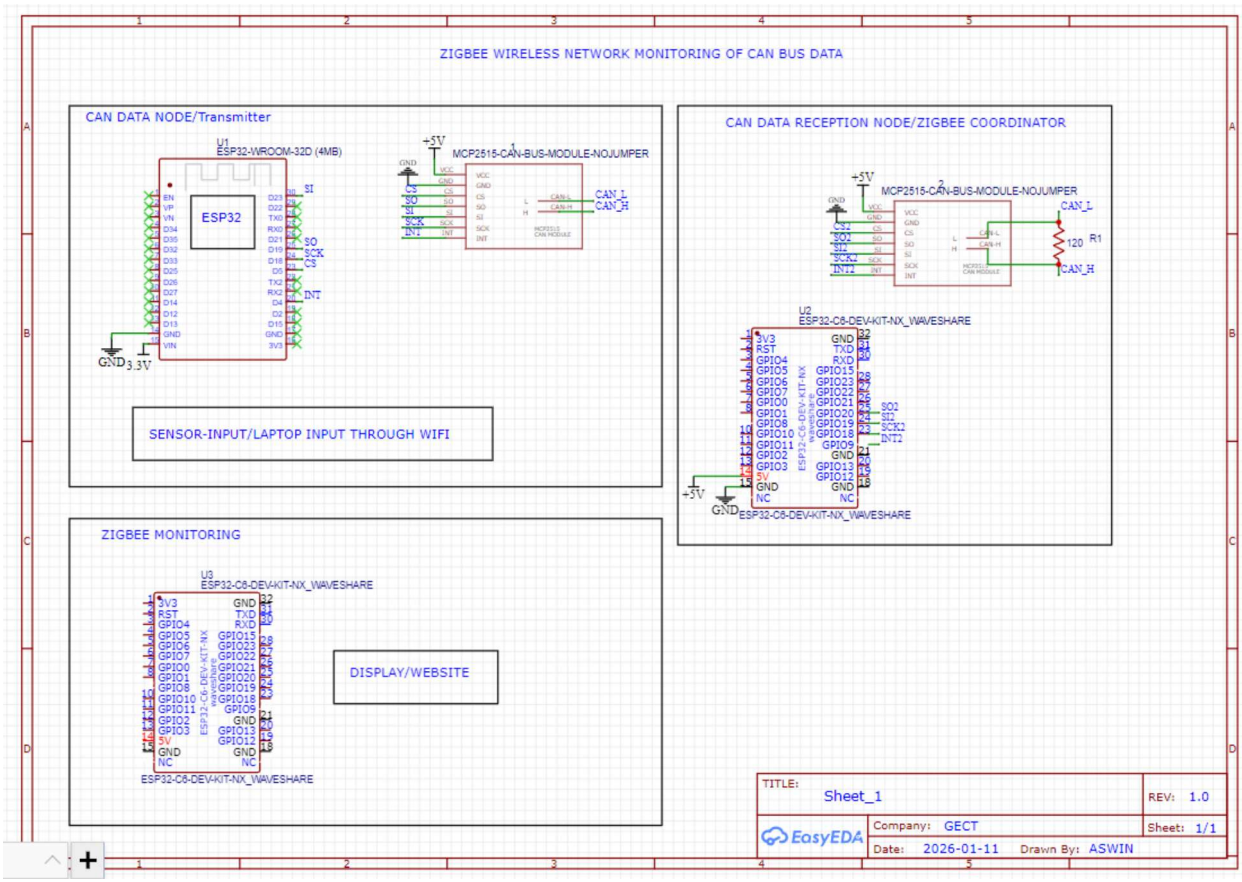


1.2 Working Principle

The system consists of sensor node connected via CAN Bus. The node collects data and transmits it over the CAN network. A central node reads the CAN data and sends it to a ZigBee module for wireless transmission to a remote monitoring station.

2 Initial Design

2.1 circuit diagram



2.2 Reason for design change

- Unavailability of esp32-h2 or esp32-c6 models
- x2c zigbee module was available in lab , so cost was reduced in a student point of view

3 New Design

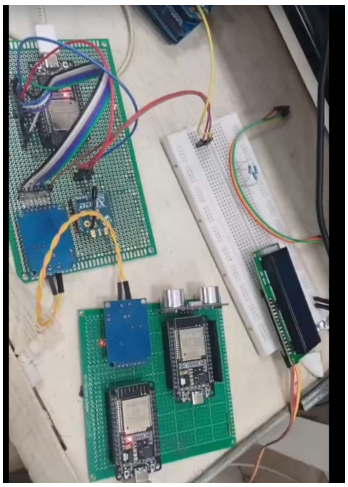


Figure 3.1: Image 1

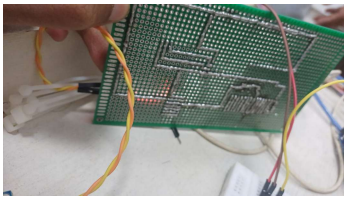


Figure 3.2: Image 2

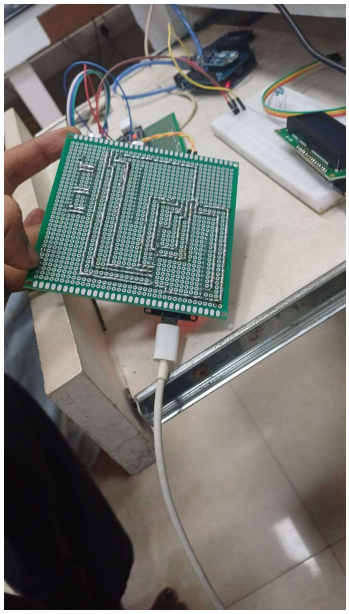


Figure 3.3: Image 3

3.1 Hardware Components

- esp32
- zigbee$_x$2c
- zigbee shield for configuring
- ultrasonic sensor
- arduino uno for configuring x2c

4 Applications

- Automotive diagnostics
- Industrial monitoring systems
- Smart grid systems
- IoT-based remote monitoring

4.1 Advantages

- Reliable communication
- Wireless monitoring capability
- Low power consumption
- Scalable system design

4.2 Limitations

- Limited ZigBee range
- Requires proper network configuration
- Initial setup complexity

4.3 Future Scope

- Integration with cloud platforms
- Use of advanced IoT protocols (MQTT)
- Mobile app-based monitoring
- AI-based predictive maintenance

4.4 Things to reflect on

- For can communication between esp32 (3.3v logic) and arduino (5v) logic donot require voltage level shifter
- Learned to use esp-idf
- understood about digikey's x2c
- It was hard to bypass security compatability of zigbee x2c and esp32-h2
- esp32h2 was not later used due to its incompatability , due to it using a different zigbee stack
- esp32 with uart x2c module was used later
- x2c has passkey of xbeealiance9 at default